

Introduction to COM31006- Computer Vision Exam

Exam

- Exam paper format
 - 1h exam
 - Close book
 - 50% of the total mark
- Please ensure that the timetable and location are in alignment with the arrangements made by the university
 - As the location is situated a distance from the city center, please allocate sufficient time for travel

Contents

- You need to answer ALL THREE Questions
 - 1 – 7 weeks, the first question
 - 8 – 11 weeks, the second and the third questions
- There are multi-choice questions
 - We indicate in the questions if it is a single-answer question or a multi-answer question
 - Partial points will be awarded for incomplete answers
- There are algorithm description questions
- There are questions about the insights of the algorithms

Examples

- We upload a practise test paper to the Blackboard

1. These questions are about Filtering, Clustering, Segmentation and Edge Detection.

- a) What is true about quantization? (single-answer) [20%]
 - (A) Affects image resolution
 - (B) Refers to the possible values used to represent a sample
 - (C) Is related to how many readings (in space) are made from a scene
 - (D) Is related to how many sensing elements the camera sensor has

- b) What is true about cross-correlation? (multiple-answer) [20%]
 - (A) Is a linear operation
 - (B) Using a vertically and horizontally flipped kernel, corresponds to convolving using the original filter
 - (C) Is useful to see how similar two signals (i.e. images) are
 - (D) Is used in the median filter

- c) Which point operator can be used for a fast increase of the low (i.e. dark) intensities of an image, and how? [20%]

- d) When using the gradient information for edge detection, what does the gradient magnitude tell us? What does the gradient direction tell us? How does the Canny operator use the gradient information? [40%]